

Database Systems: Relational, NoSQL, and Vector Stores

Relational Database Management Systems (RDBMS)

Relational databases store structured data in tables composed of predefined schemas, rows, and columns.

Relational systems guarantee transactional consistency via ACID properties:

- Atomicity: Transactions execute completely or roll back entirely without intermediate artifacts.
- Consistency: Transactions transition the database from one valid state to another, strictly enforcing constraints.
- Isolation: Concurrent transactions yield identical results as sequential execution.
- Durability: Once committed, state changes survive power failures or system crashes.

Performance optimization relies heavily on B-Tree and B+ Tree indexes, query planners, and normalization techniques to eliminate redundant data.

The NoSQL Landscape

To manage horizontal scale, high ingestion throughput, and unstructured data, NoSQL engines emerged:

- Document Stores: MongoDB and Couchbase persist semi-structured JSON/BSON records, enabling flexible, dynamic schemas.
- Key-Value Stores: Redis and AWS DynamoDB provide ultra-low sub-millisecond latency for session caching and high-velocity counters.
- Wide-Column Stores: Apache Cassandra uses consistent hashing and LSM-trees for fault-tolerant distributed multi-datacenter writes.

- Graph Databases: Neo4j models entities as nodes and relationships as edges, optimizing traversing connected social networks and fraud graphs.

With the explosion of embeddings generated by neural models, traditional lexical indexes (e.g., BM25) proved inadequate for semantic similarity. Vector databases (such as Chroma, Pinecone, Qdrant, Milvus, and pgvector) index high-dimensional floating-point vectors.

Vector stores employ Approximate Nearest Neighbor (ANN) algorithms such as Hierarchical Navigable Small World (HNSW) graphs and Inverted File with Product Quantization (IVF-PQ) to compute cosine, dot-product, or euclidean distances across millions of vectors in single-digit milliseconds, serving as the foundational retrieval backbone for Traditional and Hybrid RAG architectures.

Vector Databases for AI and Semantic Search